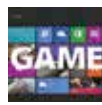




## A prediction on Internet of things

Gartner recently predicted that emerging wearable connected devices will spark off a big supply chain reaction. More here: <http://dgit.in/IOTSpark>



## Windows vs Steam?

Read the article to see what Windows has to say about Valve <http://dgit.in/1m2jMEN>

From the labs

sources like solar are being developed in the form of thermal storage. This technology relies heavily on the natural weather patterns of a region such as sunlight to generate high energy output; in Spain for example, parabolic troughs or mirrors use the focused heat from the sun to store energy in a non-toxic, non-flammable mix of sodium, potassium and calcium salts. These molten salts can be stored up to a week and used to generate electricity in a superheated steam turbine generator when electricity is needed. The Solana Generating Station in the United States is another example of one such generator in active service. It was built by the Spanish company Abengoa Solar not only as an environment friendly power station but also has a storage of 280 megawatts of energy.

The task of achieving the goal of affordable, efficient energy storage is a global one with many first world countries leading research and development. And the innovators who crack the barrier in making grid scale energy storage more efficient and affordable will not only make power distribution better but allow developing renewable sources of energy like solar and wind to become a viable alternative for large scale energy production and usage. And this breakthrough could change the way countries are placed in the struggle towards fulfilling their energy needs.

## Screenless Display

Despite the counterintuitive idea of a screenless display we have seen it conceptualised for decades, and as of now it has never been closer. The idea of a screenless display device has been seen in all forms of popular media from movies to video



Google Glass – screenless display goes mainstream

games, and with its universal utility in cell phones, cars, televisions and personal computers, scientists have already started producing a range of prototypes. These new devices could very well be the next standard device in every household.

Designed with a variety of perspectives, from projection control to hologram displays, these new devices are all part of the next wave in display technology. The most immediately accessible and perhaps most popularly known is in the form of Google Glass. Almost everyone has seen or heard of the wearable eyeglass-like computer created by Google. The technology behind Google Glass presents a layer of augmented reality over the user's sight using a small display device where the lens would be placed. The augmented reality perspective allows the display of text, information and images about the user's environment in real time as well as allows voice-activated interaction with the computer for other non-environmental tasks like email and phone calls. But this is only the first iteration of screenless display's that are in store for the future. The future is guaranteed to be stranger.

Another consumer accessible screenless display device is the Oculus Rift which has already created waves in the gaming world. The headset isolates the user from the outside world and presents visual displays from any programmed source. This form of virtual reality display takes advantage of motion tracking technologies to present an experience that is only limited by the graphics processor. The user-centric orientation combined with developing

data input devices even allow for the Rift technology being used in areas beyond gaming such as virtual robotic surgeries and air-traffic control.

The real next generation jump in screenless displays lies with innovations in holographic display technologies. Despite it's fledgling state in research and development, its potential is significant as it allows for an

interactive three dimensional image projection. Scientists are currently working with new materials and techniques to make it a reality, such as the use of optic lenses, helium neon and holographic film, which allows for a basic rudimentary holographic projection.

The most recent breakthrough came in 2013 from researchers at the Massachusetts Institute of Technology in the United States, who have already created a holographic-oriented micro-chip that can support the display of images at 50 gigapixels per second. The Object Based Media Group at MIT, that invented the e-ink technology found in e-readers like the Kindle, have used this chip to simulate real-life objects and its ability to augment projected light in multiple directions which make the use of 3D glasses redundant. Other researchers at Belgian company Imec's NVision division are utilising other approaches to the task which involves moving pixels created by reflecting lasers off microelectromechanical systems or MEMS, which are nano-components used to alter light particles.

These technologies have already excited consumer technology firms, with LG and Samsung investing millions to bring them to the users. In fact, Samsung has already filed patents in 2013 for "holographic televisions". And companies like Obscura Digital and D'strict (a design firm) have already experimented successfully with interactive projection technology using 3D projection and infrared motion sensing.

## Human Microbiome Therapeutics

Moving away from purely consumer related technologies and gadgets that



An Oculus Rift demo